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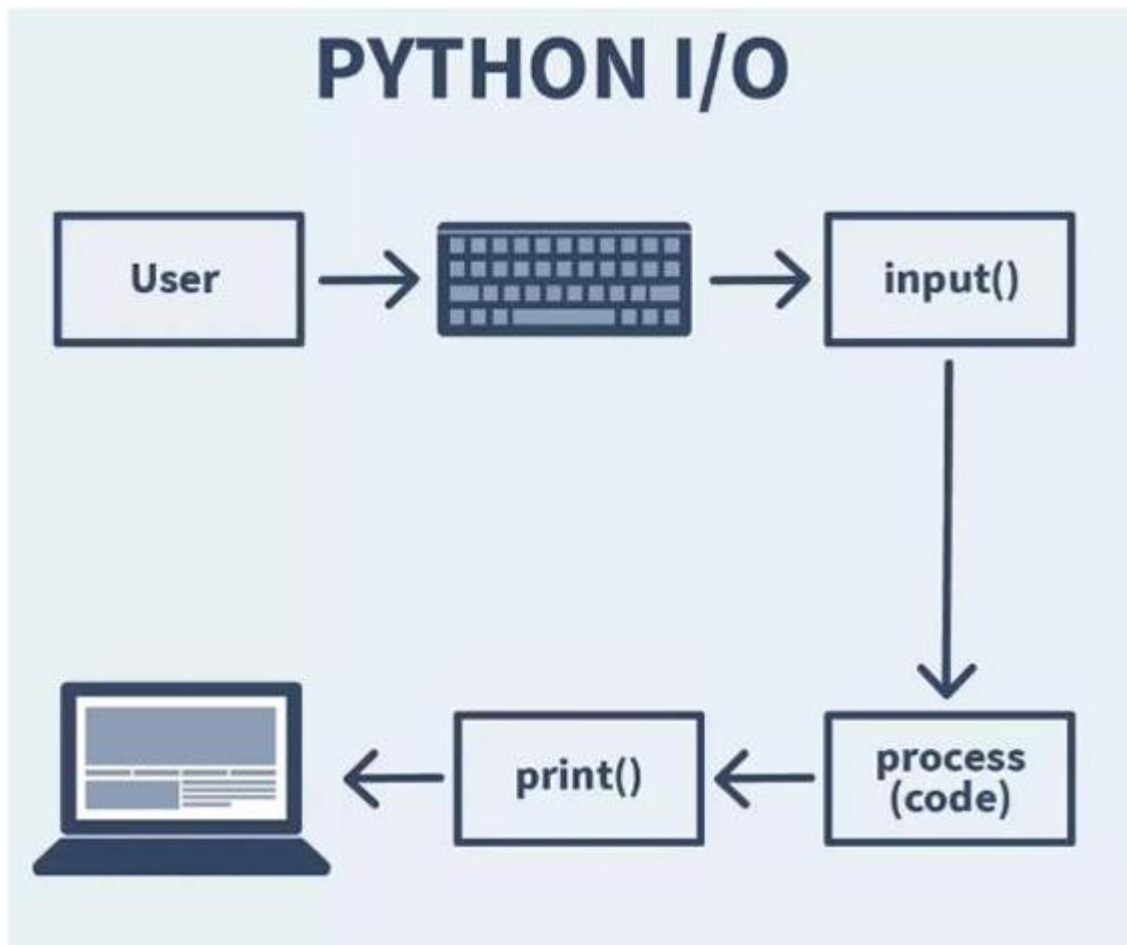
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Python Input and Output

In [Python](#) , we have many inbuilt functions that help obtain data from the user and then display the data after processing some particular logic in the code.

You can refer to the below flowchart to clarify how the system of inputting data and displaying it on some devices works.



Python Output



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- We use the widely used `print()` statement to display some data on the screen.
- We can output the particular data on some device(screen) or even in some file.
- While writing real-world programs, we have to write statements that explicitly output numbers and strings.

Let's take an example to display a simple text.

```
print("Welcome to hello world institute")
```

Don't worry much about double or single quotes. Either of them will work fine in python.

Before learning more about output in python, let's understand the actual syntax of the `print()` function. This function takes some additional arguments other than objects/values that offer more control over the output format.

Syntax:

```
print(object(s), sep=' ', end='\n', file=file, flush=False)
```

For your better understanding of the syntax here we have defined few keywords of the above statement:

- **object(s)** are the values to be printed on the screen. They are converted to strings before getting printed.



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- **sep** keyword is used to specify how to separate the objects inside the same print statement. By default, we have it as `sep=' '`, a space between two objects.
- **end** is used to print a particular thing after all the values are printed. By default, we have `end as \n`, which provides a new line after each `print()` statement.
- **file** is used to specify where to display the output. By default, it is `sys.stdout` (which is the screen).
- **flush** specifies the Boolean expression if the output is False or True. By default, it is False. In Python, the output from the `print()` goes into a buffer. Using the `flush= True` parameter helps in flushing the buffer as soon as we use the `print()` statement.

Python Output Formatting

Sometimes a user might want to make the output of a code prettier and more attractive to the audience. For this, we have some other inbuilt functions. Here we will discuss the two widely used methods.

We can use the `str.format()` method. This `format()` function will make the output more presentable.

The following curly braces `{ }` in the code acts as placeholders. We can also specify the order of these variables by putting numbers in the placeholders.

Here we take two variables and do certain operations with it.

`x = 3`



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```
y = 12
mul = x * y
print('The value of x is {} and y is {}'.format(x, y))
```

Here we are specifying the order of the variables.

```
print('{2} is the multiplication of {0} and {1}'.format(x, y, mul))
```

We can even use keyword arguments to format strings.

```
print('Hey! Welcome to {company}. In this article we will learn
about {language}'.format(company='Scaler', language='Python'))
```

Output:

The value of x is 3 and y is 12

36 is the multiplication of 3 and 12

Hey! Welcome to Scaler. In this article we will learn about Python

Besides this function, we can also use the old % method which we used similarly in C language.

```
add = 5 + 6
```

```
print("The new number after addition is %d" % add)
```

Output:

The new number after addition is 11

Python Input



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Till now, we learned about how to display the python output. The programs were not too interactive as we had hard-coded values of the variables. Sometimes users or developers want to run the programs with their own data in the variables for their own ease.

To execute this we will learn to take input from the users, in which the users themselves will define the values of the variables.

The input() statement allows us to do such things in Python.

The syntax for this function is as follows:

```
input('prompt ')
```

Here prompt is the string we want to display while taking input from the user. This is optional.

Let's look at some examples to have more clarity on this topic.

Example 1:

```
# Here we take input from the user and then display it.  
name = input("Enter your first name: ")  
print("Hey! " + name)
```

Output:

Enter your first name: Chetan

Hey! Chetan



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Example 2:

Here we will take an integer as input from the user.

```
age = int(input("Enter your age: "))
new = age + 1
name = input("Enter your name: ")
print("Hey! " + name + " Next year you will be " + str(new))
```

Output:

```
Enter your age: 19
Enter your name: Ayush
Hey! Ayush Next year you will be 20
```

Note: By default, the python input() function takes the user's input as a string. In example 2, we had to convert the age (from string to integer), which was inputted by the user, using the int() along with the input function.

We again need to convert the integer ('new' variable) into a string for concatenation during printing the output.

Example 3:

Here we will take two numbers as input from the user.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
mul = num1 * num2
print("The multiplication of num1 & num2 is: %.2f" % mul)
```



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Output:

```
Enter first number: 2.5
Enter second number: 2
The multiplication of num1 & num2 is: 5.00
```

In the print() statement we have mentioned “%.2f”. This “.2” specifies that the answer that is to be printed should be up to 2 decimal places.

Note: Instead of int(), we can also use the float() function, allowing users to enter decimal numbers instead of integers.



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